

**SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY**  
**COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT**

**Spring 2025**  
**Cloud Computing (CS-492T)**  
**Assignment # 2**

Semester: 6<sup>th</sup>  
Announced Date:03-06-25  
Total Marks: 3  
Instructor Name: Engr. Tehmeed Ebaduddin

Batch: 2022F  
Due Date:20-06-25  
Marks Obtained:

**Name: Muhammad Faisal**  
**Roll#: 2022F-BCS-152**  
**Section: A**

| CLO # | Course Learning Outcomes (CLOs)                                                                                             | PLO Mapping                                | Bloom's Taxonomy      |
|-------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-----------------------|
| CLO 2 | <b>Explain</b> system, network and storage virtualization, outline their role in enabling the cloud computing system model. | PLO_2<br>(Knowledge for Computing Problem) | C2<br>(Understanding) |

Q1. **Demonstrate** the steps to create a public load balancer to load balance VMs running Apache2 and NGINX web server using the Azure portal. Attach all the necessary snapshots of the Load balancer and the web server with IP address.

## ✓ Create Virtual Network (VNet)

Microsoft Azure

Search resources, services, and docs (G+)

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SSUET (SSUETEDU.PK)

[Home](#) > [Network foundation](#) | [Virtual networks](#) >

Create virtual network ...

Basics

Security

IP addresses

Tags

Review + create

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription \*

Azure for Students

Resource group \*

(New) Faisal152

Create new

Virtual network name \*

Faisal152-vnet

Region \* ⓘ

(US) East US

Deploy to an Azure Extended Zone

Previous

Next

Review + create

[Give feedback](#)

[Home](#) > [Network foundation](#) | [Virtual networks](#) >

Create virtual network ...

Basics

Security

IP addresses

Tags

Review + create

+ Add a subnet

10.0.0.0/16

Delete address space

10.0.0.0

/16

10.0.0.0 - 10.0.255.25565,536 addresses

| Subnets | IP address range      | Size                | NAT gateway |
|---------|-----------------------|---------------------|-------------|
| default | 10.0.0.0 - 10.0.0.255 | /24 (256 addresses) | -           |

Add IPv4 address space ⓘ

Edit subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#) ⓘ

Subnet purpose ⓘ

Default

Name \* ⓘ

default

IPv4

Include an IPv4 address space

IPv4 address range ⓘ

10.0.0.0/16

10.0.0.0 - 10.0.255.255

Starting address \* ⓘ

10.0.0.0

Size ⓘ

/24 (256 addresses)

Subnet address range ⓘ

10.0.0.0 - 10.0.0.255

IPv6

Include an IPv6 address space

This virtual network has no IPv6 address ranges.

Save

Cancel

[Give feedback](#)

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Home >

Faisal152-vnet-1751570507669 | Overview

Overview

Inputs

Outputs

Template

Search

Delete

Cancel

Redeploy

Download

Refresh

✓ Your deployment is complete

Deployment name : Faisal152-vnet-1751570507669

Subscription : Azure for Students

Resource group : Faisal152

Start time : 7/4/2025, 12:22:00 AM

Correlation ID : 7145adf4-0d46-476c-b221-c3b046186...

> Deployment details

> Next steps

Go to resource

Give feedback

Tell us about your experience with deployment

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✓ Create VM1 (Apache2)

Microsoft Azure

Search resources, services, and docs (G+)

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Home >

CreateVm-canonical.ubuntu-24\_04-lts-server-20250704004339 | Overview

Overview

Inputs

Outputs

Template

Search

Delete

Cancel

Redeploy

Download

Refresh

✓ Your deployment is complete

Deployment name : CreateVm-canonical.ubuntu-24\_04-lts-server...

Subscription : Azure for Students

Resource group : Faisal152

Start time : 7/4/2025, 12:52:23 AM

Correlation ID : 130047c9-20ee-4b45-a6d3-0089f4...

> Deployment details

> Next steps

Setup auto-shutdown Recommended

Monitor VM health, performance and network dependencies Recommended

Run a script inside the virtual machine Recommended

Go to resource

Create another VM

Give feedback

Tell us about your experience with deployment

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Home > Compute infrastructure

Compute infrastructure | Virtual machines

Overview

All resources

Infrastructure

Virtual machines

Virtual Machine Scale Set (VMSS)

Compute Fleet

Disks + images

Capacity + placement

Related services

Help

Policies

Support + Troubleshooting

Search

Virtual machines

Get started

+ Create

Switch to classic

Reservations

Manage view

Refresh

Export to CSV

Open query

Group by none

You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.

Filter for any field...

Subscription equals all

Type equals all

Resource Group equals all

Location equals all

Add filter

| Name ↑        | Subscription       | Resource Group | Location | Status  | Operating syst... | Size            | Public IP ad |
|---------------|--------------------|----------------|----------|---------|-------------------|-----------------|--------------|
| Faisal152-VM1 | Azure for Stude... | Faisal152      | East US  | Running | Linux             | Standard_D2s_v3 | 74.235.225.5 |

Showing 1 - 1 of 1. Display count: 10

Give feedback

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Microsoft Azure

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Home

Faisal152-VM1

Virtual machine

Search

Help me copy this VM in any region

Connect Start Restart Stop Hibernate Capture Delete Refresh Open in

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Connect

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Properties

Monitoring

Capabilities (7)

Recommendations

Tutorials

Virtual machine

Computer name: Faisal152-VM1

Operating system: Linux (ubuntu 24.04)

VM generation: V2

VM architecture: x64

Agent status: Ready

Agent version: 2.13.1.1

Hibernation: Disabled

Host group: -

Host: -

Proximity placement group: -

Colocation status: N/A

Capacity reservation group: -

Networking

Public IP address: 74.235.225.91 (Network interface: faisal152-vm1723)

Public IP address (IPv6): -

Private IP address: 10.0.0.4

Private IP address (IPv6): -

Virtual network/subnet: Faisal152-vnet/default

DNS name: Configure

Size

Size: Standard D2s v3

vCPUs: 2

RAM: 8 GiB

Source image details

Microsoft Azure

Search resources, services, and docs (G+)

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Home

Faisal152-VM1-ip

Public IP address

Search

Associate Dissociate Delete Move Refresh Open in mobile Give feedback

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group: Faisal152

Location: East US

Subscription: Azure for Students

Subscription ID: 580cdf0a-6029-4043-8488-ad5e336fe2f0

SKU: Standard

Tier: Regional

IP address: 74.235.225.91

DNS name: -

Domain name label s...: -

Associated to: -

Virtual machine: -

Routing preference: Microsoft network

Tags: Add tags

Get Started Properties Tutorials

Use public IP addresses for public connections to Azure resources

Associate and configure public IP addresses to various Azure resources. Learn more.

Not secure 74.235.225.91

Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/  
|-- apache2.conf  
|   |-- ports.conf  
|-- mods-enabled  
|   |-- *.load  
|   |-- *.conf  
|-- conf-enabled  
|   |-- *.conf
```

✓ Create VM2 (NGINX)

Microsoft Azure

Search resources, services, and docs (G+)

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Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Validation passed

Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Basics

Subscription

Resource group

Virtual machine name

Region

Availability options

Zone options

Security type

Enable secure boot

Enable vTPM

Integrity monitoring

Image

Azure for Students

Faisal152

Faisal152-VM2

East US

No infrastructure redundancy required

Self-selected zone

Trusted launch virtual machines

Yes

Yes

No

Ubuntu Server 24.04 LTS - Gen2

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Next >

Create

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Home >

CreateVm-canonical.ubuntu-24\_04-lts-server-20250704055419 | Overview

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name: CreateVm-canonical.ubuntu-24\_04-lts-server... Start time: 7/4/2025, 6:03:45 AM

Subscription: Azure for Students Correlation ID: 225bac84-ce5d-400d-b24d-0ac292

Resource group: Faisal152

Deployment details

Next steps

Setup auto-shutdown Recommended

Monitor VM health, performance and network dependencies Recommended

Run a script inside the virtual machine Recommended

Go to resource

Create another VM

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Home > CreateVm-canonical.ubuntu-24\_04-lts-server-20250704055419 | Overview >

Faisal152-VM2

Virtual machine

Help me copy this VM in any region

Connect

Start

Restart

Stop

Hibernate

Capture

Delete

Refresh

Open in

Essentials

Resource group (move): Faisal152

Status: Running

Location: East US

Subscription (move): Azure for Students

Subscription ID: 580cdf0a-6029-4043-8488-ad5e336fe2f0

Operating system: Linux (ubuntu 24.04)

Size: Standard D2s v3 (2 vcpus, 8 GiB memory)

Public IP address: 172.172.137.175

Virtual network/subnet: Faisal152-vnet/default

DNS name: Not configured

Health state: -

Time created: 7/4/2025, 1:04 AM UTC

Tags (edit): Add tags

Properties

Monitoring

Capabilities (7)

Recommendations

Tutorials

Virtual machine

Computer name: Faisal152-VM2

Operating system: Linux (ubuntu 24.04)

Networking

Public IP address: 172.172.137.175 (Network interface: faisal152-vm2753)

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Connect

Networking

Settings

Availability + scale

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Backup + disaster recovery

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# Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to [nginx.org](http://nginx.org).  
Commercial support is available at [nginx.com](http://nginx.com).

*Thank you for using nginx.*

## ✓ Create Azure Load Balancer

Microsoft Azure

Search resources, services, and docs (G+)

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Home > Load balancing and content delivery | Load balancers >

Create load balancer

Subscription \*

Azure for Students

Resource group \*

Faisal152

Create new

Instance details

Name \*

Faisal152-LB

Region \*

East US

SKU \*

☒ Standard (Distribute traffic to backend resources)

☐ Gateway (Direct traffic to network virtual appliances)

Type \*

☒ Public

☐ Internal

Tier \*

☒ Regional

☐ Global

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Review + create

< Previous

Next: Frontend IP configuration >

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## ✓ Add frontend IP configuration

Microsoft Azure

Search resources, services, and docs (G+)

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Home > Load balancing and content delivery > Load balancers >

Create load balancer

BasicsFrontend IP configurationBackend poolsInbound rulesOutbound rulesTagsReview + create

A frontend IP configuration is an IP address used for inbound and/or outbound communication as defined within load balancing, in

+ Add a frontend IP configuration

| Name ↑↓                          | IP address ↑↓ |
|----------------------------------|---------------|
| Add a frontend IP to get started |               |

Review + createPreviousNext : Backend pools >Download a template for automationGive feedback

Add frontend IP configuration

Faisal152-LB

Name \*

Faisal152-FE-IP-config

IP version

☒ IPv4

☐ IPv6

Add a public IP address

Name \*

Faisal152-Public-IP

SKU

Standard

Tier

Regional

Static IPs are assigned at the time the resource is created and released when the resource is deleted. Dynamic IPs are assigned when associating the IP to a resource and is released when you stop, restart, or delete a resource. Dynamic is only available for Basic SKU.

Assignment

☐ Dynamic

☒ Static

Availability zone \*

Zone-redundant

Routing preference ⓘ

☒ Microsoft network

☐ Internet

Save

Cancel

✓ Add Backend Pool

Microsoft Azure

Search resources, services, and docs (G+/)

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Home > Load balancing and content delivery > Load balancers > Create load balancer >

Add backend pool

Name \*

Faisal152-BE-Pool

Virtual network ⓘ

Faisal152-vnet (Faisal152)

Backend Pool Configuration

NIC

IP address

⚠ When a backend pool is configured by IP address, the backend instances are not secure by default and still use default outbound access. To secure your backend pool, please use a private subnet parameter. [Learn more](#)

IP addresses

You can only add resources IP address in the Virtual Network. The configuration is associated with the IP address and will apply to any resource which has this IP address assigned.

| Backend Address Name             | IP address | Resource Name             |
|----------------------------------|------------|---------------------------|
| d7418aae-c3c2-421b-b3c6-875fe... | 10.0.0.4   | Faisal152-VM1 (Faisal152) |
| 712095a3-392c-44a1-b3a1-631a...  | 10.0.0.5   | Faisal152-VM2 (Faisal152) |

Save

Cancel

Give feedback

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Home > Load balancing and content delivery > Load balancers >

Create load balancer

Basics

Frontend IP configuration

Backend pools

Inbound rules

Outbound rules

Tags

Review + create

A backend pool is a collection of resources to which your load balancer can send traffic. A backend pool can contain virtual machines, virtual machine scale sets, and containers.

+ Add a backend pool

| Name              | Virtual network | Resource Name | Network interface | IP address | Availability zone | Admin state |
|-------------------|-----------------|---------------|-------------------|------------|-------------------|-------------|
| Faisal152-BE-Pool |                 |               |                   |            |                   |             |
| Faisal152-BE-Pool | Faisal152-vnet  | Faisal152-VM1 | faisal152-vm1723  | 10.0.0.4   | -                 |             |
| Faisal152-BE-Pool | Faisal152-vnet  | Faisal152-VM2 | faisal152-vm2753  | 10.0.0.5   | -                 |             |

Review + create

< Previous

Next : Inbound rules >

Download a template for automation

Give feedback

✓ Add Load Balancing Rule

Microsoft Azure

Search resources, services, and docs (G+/)

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Home > Load balancing and content delivery > Load balancers >

Create load balancer

Basics

Frontend IP configuration

Backend pools

Inbound rules

Outbound rules

Tags

Review + create

Load balancing rule

A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend instances that are eligible to receive traffic.

+ Add a load balancing rule

| Name ↑↓                   | Frontend IP configuration ↑↓ | Backend pool ↑↓ | Health probe ↑↓ |
|---------------------------|------------------------------|-----------------|-----------------|
| Add a rule to get started |                              |                 |                 |

Inbound NAT rule

An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.

+ Add an inbound nat rule

| Name ↑↓                   | Frontend IP configuration ↑↓ | Service ↑↓ |
|---------------------------|------------------------------|------------|
| Add a rule to get started |                              |            |

Review + create

< Previous

Next : Outbound rule >

Download a template for automation

Give feedback

Add load balancing rule

Faisal152-LB

Name \*

Faisal-152-LB-Rule

IP version \*

IPv4

IPv6

Frontend IP address \* ⓘ

Faisal152-FE-IP-config (To be created)

Backend pool \* ⓘ

Faisal152-BE-Pool

Protocol

TCP

UDP

Port \*

80

Backend port \* ⓘ

80

Health probe \* ⓘ

No existing probes

Save

Cancel

Give feedback

✓ Create Health-Probe

Microsoft Azure

Search resources, services, and docs (G+/I)

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Home > Load balancing and content delivery | Load balancers >

Create load balancer

Basics Frontend IP configuration Backend pools Inbound rules Outbound rules Tags Review + create

Load balancing rule

A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend instances that are eligible to receive traffic.

+ Add a load balancing rule

Name ↑↓ Frontend IP configuration ↑↓ Backend pool ↑↓ Health probe ↑↓

Add a rule to get started

Inbound NAT rule

An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.

+ Add an inbound nat rule

Name ↑↓ Frontend IP configuration ↑↓ Service ↑↓

Review + create < Previous Next : Outbound rule > Download a template for automation Give feedback

Add load balancing rule

Faisal152-LB

UDP

Port \* 80

Backend port \* 80

Health probe \* No existing probes

Health probes are used to check the status of a backend pool instance. If the health probe fails to get a response from a backend instance then no new connections will be sent to that backend instance until the health probe succeeds again.

Name \* Faisal152-Health-probe

Protocol \* TCP

Port \* 80

Interval (seconds) \* 5

Used by \* Not used

Save Cancel

Microsoft Azure

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SSUET

Home > Load balancing and content delivery | Load balancers >

Create load balancer

Basics Frontend IP configuration Backend pools Inbound rules Outbound rules Tags Review + create

Load balancing rule

A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. The load balancing rule uses the health probe to check the status of the backend pool instances.

+ Add a load balancing rule

| Name ↑↓            | Frontend IP configuration ↑↓ | Backend pool ↑↓   | Health probe ↑↓        | Frontend Port ↑↓ | Backend port ↑↓ |
|--------------------|------------------------------|-------------------|------------------------|------------------|-----------------|
| Faisal-152-LB-Rule | Faisal152-FE-IP-config       | Faisal152-BE-Pool | Faisal152-Health-Probe | 80               | 80              |

Inbound NAT rule

An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.

+ Add an inbound nat rule

Name ↑↓ Frontend IP configuration ↑↓ Service ↑↓ Target ↑↓ Frontend Port ↑↓

Review + create < Previous Next : Outbound rule > Download a template for automation Give feedback

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SSUET

Home >

Microsoft.LoadBalancer-20250704060837 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name : Microsoft.LoadBalancer-202507040608... Start time : 7/4/2025, 6:23:57 AM

Subscription : Azure for Students Correlation ID : b94e780c-96df-410a-b539-19fe4e528a...

Resource group : Faisal152

Deployment details

Next steps

Go to resource

Give feedback

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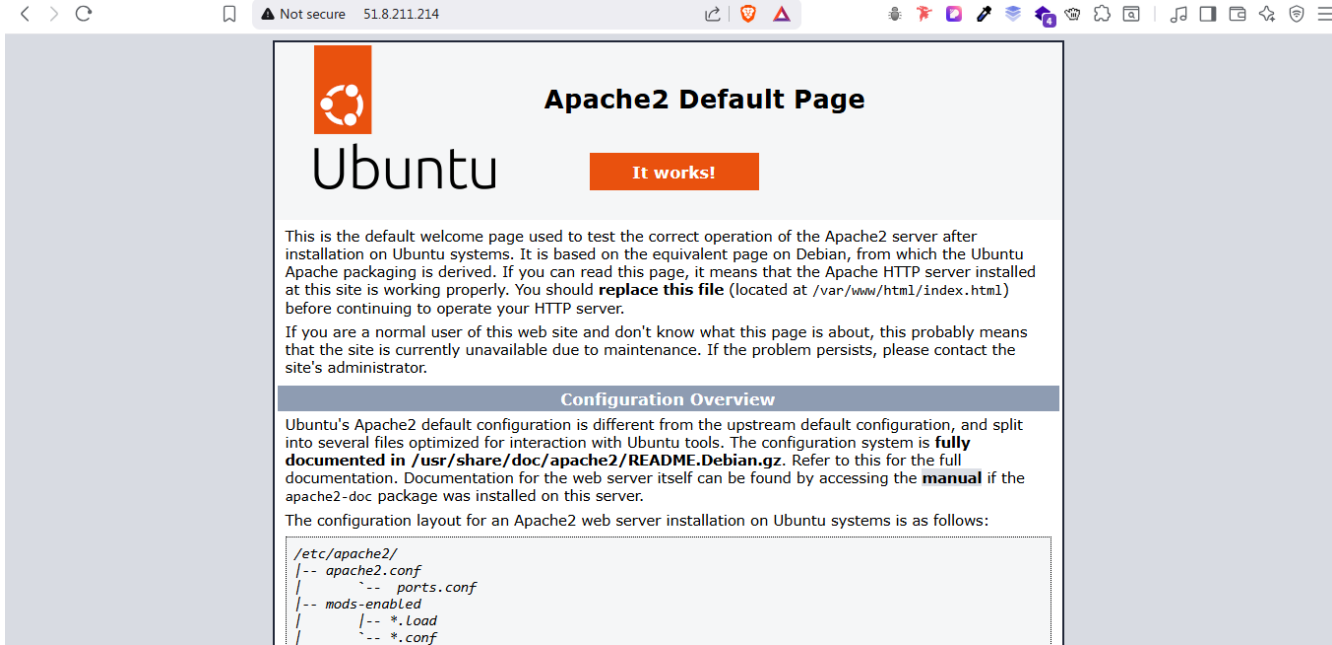
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✓ **Dissociate VM public IP on Azure**

